

2. **apt-get**: A more advanced package management system, it makes use of the topological sorting technique to resolve dependencies of the packages, and calls *dpkg* at the appropriate times for installation or deletion of packages. It uses archives of Ubuntu and Canonical as repositories. The command...

`sudo apt-get install <package-name>`

...will install the package. *sudo* is used since package installation requires administrative rights.

3. **Aptitude**: This provides a graphical interface for *apt*.
4. **pip**: *pip* stands for Pip Installs Packages. It is a Python package manager. Any Python package available in the PyPI repository can be installed using *pip*. The command for installing this package is:

`pip install <package-name>`

Though package managers like *pip* can deal with almost all Python packages, they neglect non-Python packages, which are dependencies of our package of interest.

Package dependencies may differ because package managers differ. This is because different package managers refer to different repositories. Some repositories may contain a lot of packages, and package managers referring to these repositories will install the necessary packages and all the dependencies in that repository.

Figure 1, Figure 2 and Figure 3 illustrate the scenario when I tried to install IPython using *apt-get*, *conda* and *pip*. We can see that the list of dependencies is different. Just remember that when more packages are used in a program, they may occupy more space.

Anaconda and Miniconda

A distribution is a collection of pre-compiled and pre-configured packages that work together. Anaconda and Miniconda are free Python distributions that provide both an environment manager and a package manager. They are helpful when you are into deep learning or for data science applications, but are not limited to these domains. Other than the package manager Conda, Conda-build, Python and over 150 packages are automatically downloaded with Anaconda. Miniconda includes Conda, Python and its dependencies only.

The power of Anaconda can be explained with the help of the following example. If you are into deep learning, you may need many packages. Applications using ResNet may require packages like *pillow*, *Keras* and *Theano* for proper working. If you install Anaconda, all these packages are automatically installed. Installation of Miniconda may require us to install many packages as and when required.

According to Continuum Analytics, Conda is an open source environment management system and package

```
inspiron-N5110:/# sudo apt-get install ipython
[sudo] password for surrykautan:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  python-decorator python-simplegeneric
Suggested packages:
  python-doc ipython-notebook ipython-gtkconsole python-matplotlib python-zmq
The following NEW packages will be installed:
  ipython python-decorator python-simplegeneric
0 upgraded, 3 newly installed, 0 to remove and 7 not upgraded.
Need to get 631 kB of archives.
After this operation, 3,461 kB of additional disk space will be used.
Do you want to continue? [y/n]
```

Figure 1: IPython using *apt-get*

```
inspiron-N5110:/# conda install ipython
Fetching package metadata .....
Solving package specifications: .
Package plan for installation in environment /home/inspiron-N5110/miniconda2:

The following NEW packages will be INSTALLED:

backports: 1.0-py27_0
decorator: 4.0.11-py27_0
get_terminal_size: 1.0.0-py27_0
ipython: 5.3.0-py27_0
ipython_genutils: 0.1.0-py27_0
path.py: 10.1-py27_0
pathlib2: 2.2.0-py27_0
pexpect: 4.2.1-py27_0
pickleshare: 0.7.4-py27_0
prompt_toolkit: 1.0.15-py27_0
ptyprocess: 0.5.1-py27_0
pygments: 2.2.0-py27_0
scandir: 1.5-py27_0
simplegeneric: 0.8.1-py27_1
traitlets: 4.3.2-py27_0
wcwidth: 0.1.7-py27_0

Proceed [y/n]? [y]
```

Figure 2: IPython using *conda*

```
The following NEW packages will be INSTALLED:

backports: 1.0-py27_0
decorator: 4.0.11-py27_0
enum34: 1.1.6-py27_0
get_terminal_size: 1.0.0-py27_0
ipython: 5.3.0-py27_0
ipython_genutils: 0.2.0-py27_0
openssl: 1.0.2k-1
path.py: 10.3.1-py27_0
pathlib2: 2.2.1-py27_0
pexpect: 4.2.1-py27_0
pickleshare: 0.7.4-py27_0
pip: 9.0.1-py27_1
prompt_toolkit: 1.0.14-py27_0
ptyprocess: 0.5.1-py27_0
pygments: 2.2.0-py27_0
python: 2.7.13-0
readline: 6.2-2
scandir: 1.5-py27_0
setuptools: 27.2.0-py27_0
simplegeneric: 0.8.1-py27_1
six: 1.10.0-py27_0
sqlite: 3.13.0-9
tk: 8.5.18-0
traitlets: 4.3.2-py27_0
wcwidth: 0.1.7-py27_0
wheel: 0.29.0-py27_0
zlib: 1.2.8-3

Proceed [y/n]? [y]
```

Figure 3: IPython dependencies when installed using *pip*

management system for installing multiple versions of packages and its dependencies. It was first developed in 2012.

Conda installation

Conda can be installed along with Anaconda or Miniconda. You can also purchase an Anaconda subscription to install Conda. The choice between Anaconda and Miniconda depends on the time available and the disk space. If you have minimum 3GB disk space to spare, and need all the options of Conda, download Anaconda. If you have low disk space and you just need a start with Conda, a minimal version of Anaconda called Miniconda will be the best option, as it takes only 400MB of space. You can download a 32-bit or 64-bit